DESIREE MANICARDI

Ph.D. in *Computer Science and Computational Mathematics*

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SUMMARY

- Bachelor's and Master's degrees in *Computer Science* (CS) + Ph.D. in *Computer Science and Computational Mathematics*.
- 4+ years of experience in academia, focusing on *modelling biological systems* in ***Python*** – code available on dmanicardi
- 15+ projects at university (further information here): 8 considering sw requirements and UML, 2 using JavaScript and 2 using SQL.
- 20+ attended CS courses in the past 4 years, while several courses in *computer engineering* and *requirements analysis* during my Master's degree.
- Since December 2023, I have been managing my website using *HTML*, *CSS*, *Bootstrap*, and *JavaScript*.

- Italian: native.
- **English**: *Upper-Intermediate level*: FCE Certificate (CEFR B2) & PhD in English & Scientific Papers & Speaker @ ICTCS 2023.

CS SKILLS *selected*

Mainly used:
Java | Python | SQL | NoSQL

Already used:
Programming: C; C++; C#
Programming technique: OOP; real-time
Web: HTML; PHP; CSS; Bootstrap; JavaScript; AngularJS; AJAX
Database: PostgreSQL; MySQL; PHPMyAdmin; MongoDB
Diagrams : ER; UML; KAOS; Problem Frames
Formats : JSON
IDEs & platforms : Eclipse; Node.js; Android Studio; ArduinoUNO
Tools: pgAdmin; Node-RED
VCS: GitHub
SO: Linux; Windows

HOBBY

Genealogy Tree (already since 1768) & Learning Turkish (CEFR A1-A2)

FOR FURTHER INFORMATION

See my my website or scan the QR code:

EDUCATION

Ph.D. in *Computer Science and Computational Mathematics* | University of Insubria

11 2020 – 05 2024 | Como, Italy

- Thesis: *Formal Models for Biological Systems*.
- I focused my research on *formal models for biological systems*. I mainly worked on a model to formalise a process algebra. It is user-friendly, easy to implement, and tailored for modelling *biochemical reaction networks*. The main contribution is related to the study of the *robustness* of biochemical networks.
- Main courses (in English): *Bioinformatics*, *IoT*, *Cryptography*, *Modeling and Verifying Hybrid Systems*.
- Camp: *Innovation Camp for PhD Students – Deep Dive Into Innovation and Execution* (03-05/2021, in English).

Master's degree in *Computer Science* | University of Insubria

12 2016 – 07 2018 | Varese, Italy

- Thesis: *Modelling a Dual Chamber Pacemaker in Span(Graph)* (in Italian), 110/110 cum laude.
- Main courses: *Software Engineering*, *Requirements Engineering*, *Programming in Java and C*, *Models of Computation*, *Models for Biological Systems*, *Privacy and Security*, *Real-time programming*, *Artificial Intelligence*.

Bachelor's degree in *Computer Science* | University of Insubria

09 2013 – 12 2016 | Varese, Italy

- Thesis: *Design and development of an application for filtering Twitter messages*, 101/110.
- Main courses: *Database*, *Privacy and Security*, *Coding theory*, *Programming in Java*, *Software Design*, *Web applications*.

WORKING EXPERIENCE

Didactic tutor for the courses on *Didattica dell'Informatica* | Università Milano-Bicocca

04 2025 – 12 2025 | Milano

Recognised Researcher | University of Insubria

03 2024 – 02 2025 | Como, Italy

- Research Title: *Specification and analysis of behavioural differences in biomedical systems*.
- PRIN 2022 Project: *MEDICA: Modelling and vErification of alkaptonuria and multiple sclerosis Driven by biomedICAl data*.
- Programming languages: *Python* and mainly *Java*.
- See the website *here*.

Traineeships | University of Insubria

- (11 2017 – 05 2018) Analysis of the Span-Cospan(Graph) formalism, focusing on timing.
- (02 2016 – 09 2016) Design and development of a web application aimed at filtering tweets using Twitter API.

TEACHING EXPERIENCE

Experience in *delivering* ***Python*** *and* ***Java*** *laboratory lectures* for Bachelor's and Master's students, in **Italian**.